

ACCESS KNOB: PARAMETRIC DESIGN OF 3D-PRINTABLE ACCESSIBLE SYNTHESIZER CONTROLS

Dillon Simeone¹, Shawn Trail², Doga Cavdir^{3,4}

¹Universal Music Design / CymaSpace (Portland, OR) • ²Confluence Recording Co. (Portland, OR) • ³IT University of Copenhagen • ⁴Stanford University (CCRMA)

ASSETS '26

The 28th International ACM SIGACCESS Conference
Athens, Greece • October 26-28, 2026

INTRODUCTION & CO-DESIGN CONTEXT

Commercial synthesizer interfaces present significant accessibility barriers for musicians with motor and sensory impairments. Standard controls, which are typically small, smooth, and closely spaced, require a high-precision pinch grip and considerable physical force to actuate.

"Dialysis-dependent kidney failure caused severe hand neuropathy in a retired IT worker and synth collector, leaving his collection of synthesizers completely unplayable."

This lived encounter inspired **"ACCESS KNOB"**, an open-source browser-based parametric customizer. Built on Universal Music Design (UMD) principles, the project shifts the paradigm from mass-produced controls to personalized, user-fabricated interfaces that enhance musical expression and independent control.

PROCEDURAL WASM CAD KERNEL

Traditional CAD tools present steep learning curves and lack accessibility. ACCESS KNOB utilizes a client-side WebAssembly-compiled **"Manifold"** geometry kernel to perform boolean operations in milliseconds, enabling real-time parameter tweaking directly in the browser.

Two Mounting Modes & Three Spindle Bores:

- Swap-In Mode:** Replaces the original manufacturer knob entirely. Generates custom internal flat/splined bores to slide flush onto the bare metal/plastic spindle.
- Slide-Over Mode:** An adaptive sleeve that slides snugly over the original control cap, preserving the instrument's resale value and avoiding complex hardware disassembly.
- Spindle Bore Profiles:** Generates custom flats/splines for **D-Shafts**, **Knurled/Splined** (18/24-tooth gears), and **Solid Round** spindles (secured via integrated M3 set-screw).



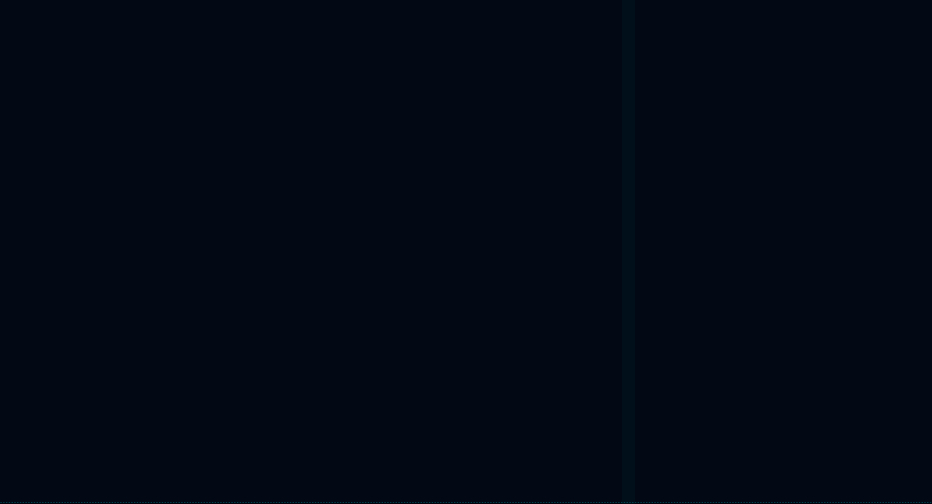
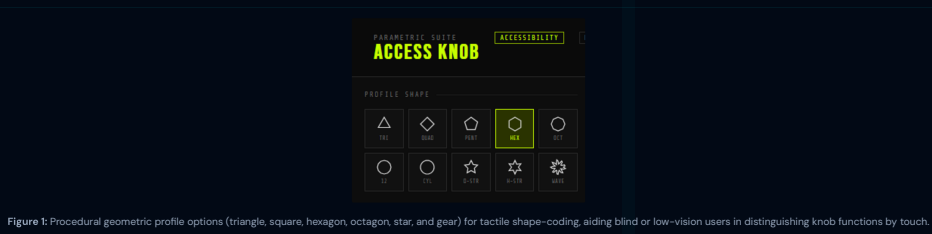
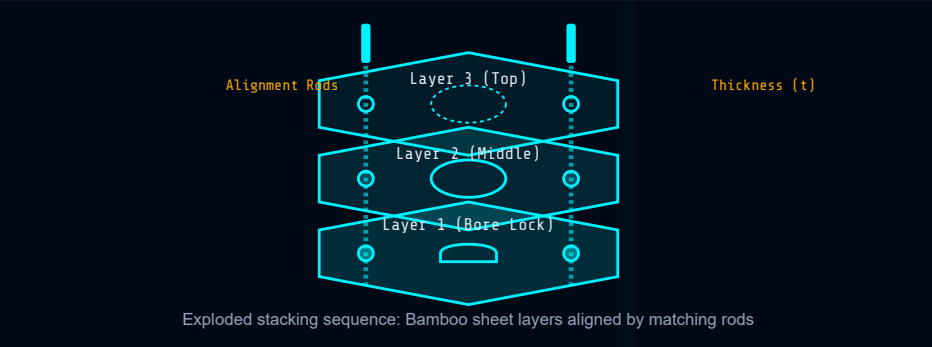
2D VECTOR SVG STACKING EXPORTER

To democratize fabrication for makerspaces without access to 3D printers, ACCESS KNOB includes a 2D vector SVG stacking exporter. This tool enables subtractive manufacturing (laser-cutting or CNC routing) using sheet materials (wood, acrylic, bamboo, or composites).

"The exporter dynamically slices the 3D model into layer profiles, compensates for laser kerf, and adds kerf-compensated slot paths and stacking rods."

System Assembly Mechanics:

- Kerf Compensation:** Sub-millimeter adjustment (default 0.1 mm) ensures slots and outer boundaries match target shapes after laser-cutting.
- Dual Alignment Rods:** Generates two rods of width thickness and length heightKnob + kerf. Slits are cut through all stacked layers, allowing the rods to slide through, aligning the pieces perfectly for gluing.
- Layer Taper:** Outer contours are calculated iteratively at height intervals to preserve the overall tapered ergonomic profile.



TACTILE MATERIAL DYNAMICS

Material choice impacts physical sensation, thermal stability, and control responsiveness during musical play:

Material	Process	Rigidity	Friction	Tactile Feedback & Damping
PLA	FDM 3D Print	High	Medium	Crisp edges; low vibrational damping
PETG	FDM 3D Print	Medium	Medium	Slight elasticity (ideal for outer sleeves)
TPU	FDM 3D Print	Very Low	High	Flexible; excellent non-slip inner lining
Bamboo	Laser-Cut Stack	High	High (Grain)	Organic feel; superior mechanical damping

MECHANICAL EVALUATION

To evaluate how parametric adjustments reduce required physical force, we model the peripheral finger force F required to rotate a potentiometer with torque resistance T :

$$T = F \cdot (D / 2) \Rightarrow F = 2T / D$$

Increasing the outer diameter D from a standard 10 mm to an accessible 32 mm or 40 mm reduces the required finger pinch force by **68.8% to 75.0%**, compensating directly for reduced grip strength.

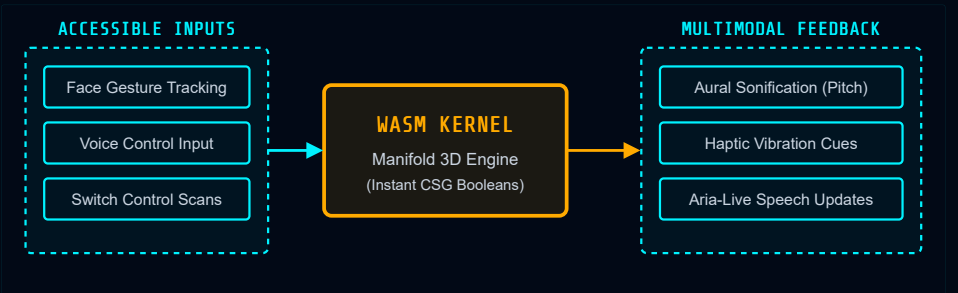
Diameter (D)	Required Force (F)	Force Reduction
10 mm (Standard)	$F_{10} = 2T / 10$	0.0% (Baseline)
15 mm	$0.67 \cdot F_{10}$	33.3%
25 mm (Medium)	$0.40 \cdot F_{10}$	60.0%
32 mm (Large)	$0.31 \cdot F_{10}$	68.8%
40 mm (Accessible)	$0.25 \cdot F_{10}$	75.0%

Additionally, increasing height H expands the usable vertical contact area $A = \pi \cdot D \cdot H$, allowing palm-dragging or side-of-hand shear techniques that eliminate the need for a precise pinch grip.

ACCESSIBILITY & FUTURE WORK

Hands-Free Design Customization:

The web customizer features high-contrast glowing visual focus states, supporting commodity webcam head-tracking and facial gesture interfaces (Google Project Gameface, Apple Voice/Switch Control).



Future Work & Clinical Evaluation:

We are expanding the customizer to generate fader caps and drum pads. We will conduct an IRB-approved clinical evaluation with $N = 12$ musicians with hand impairments to assess long-term ergonomic efficacy, task speeds, and subjective creative agency.